

Salt Marsh Vegetation at NCOS

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Guiding questions

- Which dominant plant species characterize salt marsh habitat at NCOS? (figure 3, showing the full species rank and identifying which species have the highest mean cover overall)
- How has the mean percent cover of the dominant salt marsh species changed across survey years? (figure 1, showing top species' cover through time)
- How has the relative contribution of native and non-native vegetation changed across survey years? (figure 2, showing native vs. non-native cover through time)

1. Set up

Packages

```
# general use  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.3      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
# file organization
library(here)
```

here() starts at /Users/ej/GitHub/ncos-vegetation

```
# data visualization
library(patchwork)
library(naniar)
```

Data

```
veg <- read_csv(here("data", "veg.csv"))
```

Rows: 27036 Columns: 17

-- Column specification -----

Delimiter: ","

chr (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...

dbl (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Rows: 257 Columns: 4

-- Column specification -----

Delimiter: ","

chr (2): year_pool, transect_name

dbl (2): year, num_quad

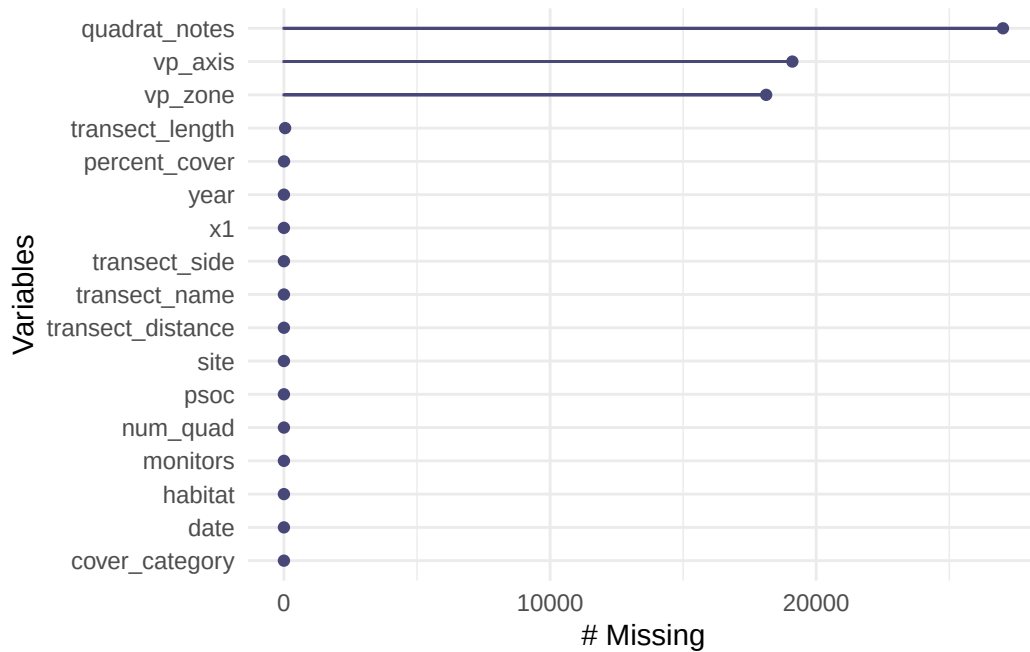
i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

2. Cleaning and wrangling

Explore missingness

```
gg_miss_var(veg)
```



Filter to salt marsh habitat and late summer/early fall surveys

```
saltmarsh <- veg |>
  # filter to salt marsh habitat only
  filter(str_detect(habitat, "Salt Marsh") | str_detect(habitat, "Brackish Marsh")) |>
  # parse date column and filter to August and September
  mutate(date = as.Date(date),
         month = month(date)) |>
  filter(month %in% c(8, 9))
```

Fill implicit NAs with 0

```
saltmarsh_complete <- saltmarsh |>
  complete(year, transect_name, psoc, transect_distance,
           fill = list(percent_cover = 0)) |>
  group_by(year, transect_name, psoc, cover_category) |>
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  ungroup() |>
  unite("year_pool", year, transect_name, remove = FALSE) |>
  left_join(metadata, by = "year_pool") |>
  select(-year.y, -transect_name.y) |>
  rename(year = year.x,
         transect_name = transect_name.x) |>
  mutate(mean_pc = sum_pc / num_quad)
```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by year, transect_name, psoc, and cover_category.
- i Output is grouped by year, transect_name, and psoc.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(year, transect\_name, psoc, cover\_category))` for per-operation grouping (`dplyr::dplyr\_by`) instead.

Filter to plant cover categories only

```
saltmarsh_plants <- saltmarsh_complete |>
  filter(cover_category %in% c("NATIVE COVER", "NON-NATIVE COVER"))
```

3. Summarizing for each figure

Figure 3: Species rank (averaged across all years)

```
# summarize-species-rank - keep SE this time
species_rank <- saltmarsh_plants |>
  group_by(psoc, cover_category) |>
  summarize(mean_pc = mean(mean_pc, na.rm = TRUE),
```

```

      se_pc = sd(mean_pc, na.rm = TRUE) / sqrt(n()),
      n = n()) |>
ungroup() |>
arrange(desc(mean_pc))

```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by psoc and cover_category.
- i Output is grouped by psoc.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(psoc, cover\_category))` for per-operation grouping (`?dplyr::dplyr\_by`) instead.

```

species_rank_top20 <- species_rank |>
  filter(n > 1) |>
  slice_max(mean_pc, n = 20)

```

Figure 1: Dominant species by year

```

# identify top 5 dominant species from species_rank
dominant_species <- species_rank |>
  slice_max(mean_pc, n = 5) |>
  pull(psoc)

# summarize dominant species cover by year - raw mean percent cover
dominant_by_year <- saltmarsh_plants |>
  # label dominant species; group rest as Other
  mutate(psoc_grouped = if_else(psoc %in% dominant_species, psoc, "Other")) |>
  # filter to only dominant species
  filter(psoc_grouped != "Other") |>
  # get transect-level totals per species per year first
  group_by(year, transect_name, psoc_grouped, cover_category) |>
  summarize(sum_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  # then average across transects within each year and species
  group_by(year, psoc_grouped, cover_category) |>
  summarize(mean_pc = mean(sum_pc, na.rm = TRUE)) |>
  ungroup()

```

`summarise()` has regrouped the output.

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by year, transect_name, psoc_grouped, and
  cover_category.
i Output is grouped by year, transect_name, and psoc_grouped.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(year, transect_name, psoc_grouped, cover_category))`
  for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

Figure 2: Native/non-native proportions by year

```
native_nonnative <- saltmarsh_plants |>
  group_by(year, cover_category) |>
  summarize(mean_pc = mean(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  group_by(year) |>
  mutate(prop = mean_pc / sum(mean_pc)) |>
  ungroup()
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by year and cover_category.
i Output is grouped by year.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(year, cover_category))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Figure 4: Total vegetation cover by year

```
total_cover <- saltmarsh_plants |>
  group_by(year, transect_name) |>
  summarize(total_pc = sum(mean_pc, na.rm = TRUE)) |>
  ungroup() |>
  group_by(year) |>
  summarize(mean_total = mean(total_pc, na.rm = TRUE),
            se_total = sd(total_pc, na.rm = TRUE) / sqrt(n())) |>
  ungroup()
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by year and transect_name.
```

- i Output is grouped by year.
- i Use ``summarise(.groups = "drop_last")`` to silence this message.
- i Use ``summarise(.by = c(year, transect_name))`` for per-operation grouping (``?dplyr::dplyr_by``) instead.

4. Visualizations

Figure 1: Dominant species percent cover across survey years

```
fig1 <- ggplot(data = dominant_by_year,
               aes(x = year,
                   y = mean_pc,
                   fill = psoc_grouped)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c(
    "Salicornia_pacifica"      = "#0D5E45",
    "Distichlis_spicata"      = "#1D9E75",
    "Atriplex_lentiformis"    = "#5DCAA5",
    "Eleocharis_macrostachya" = "#A8DBC9",
    "Raphanus_sativus"       = "#D85A30"
  )) +
  scale_x_continuous(breaks = seq(2018, 2025, 1)) +
  labs(x = "Survey year",
       y = "Mean percent cover (%)",
       fill = "Species") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(face = "italic"),
        legend.title = element_text(face = "bold"))

fig1
```

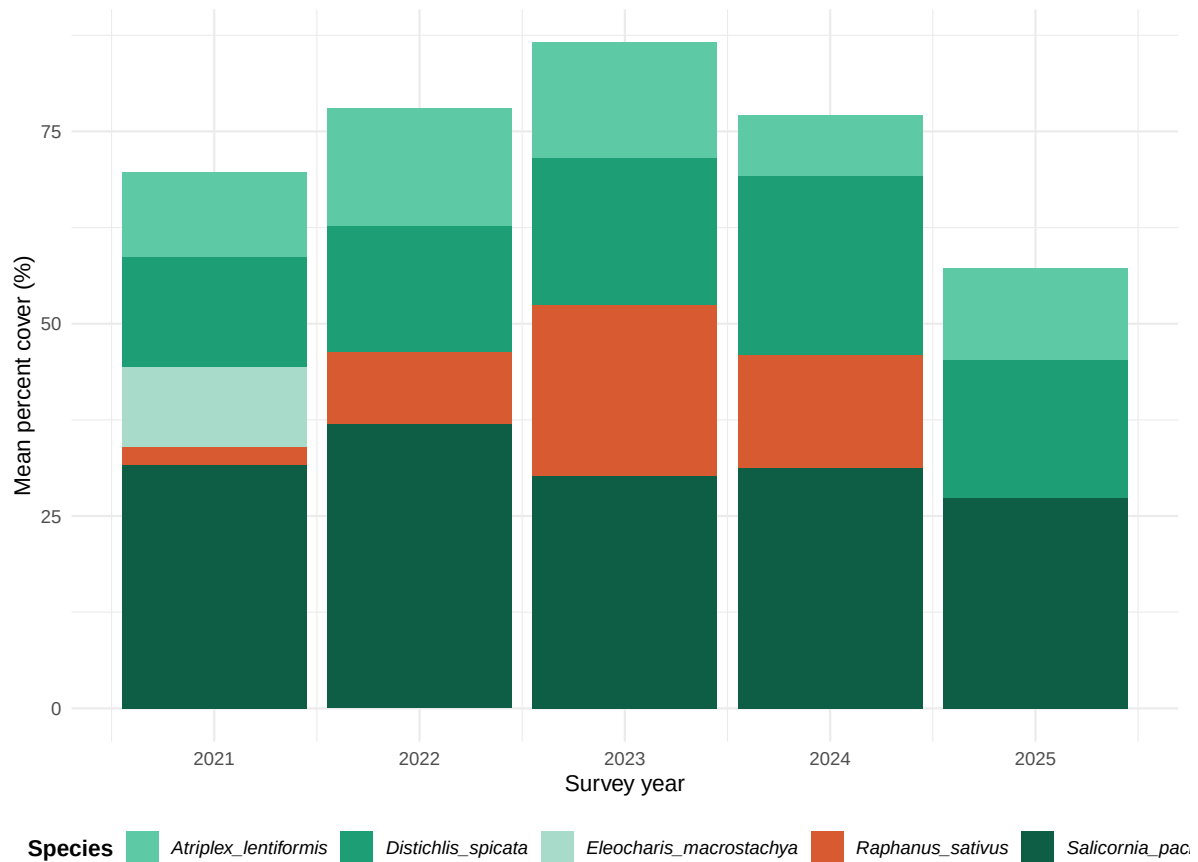


Figure 1: Mean percent cover of the five dominant salt marsh plant species at NCOS across survey years, averaged across transects within each year. Dominant species were identified as the top five by mean percent cover across all survey years. Bar height reflects total mean cover of the five dominant species and varies across years. Green shades indicate native species; coral indicates non-native species.

Figure 2: Proportional native vs. non-native cover across survey years

```
fig2 <- ggplot(data = native_nonnative,
               aes(x = year,
                   y = prop,
                   fill = cover_category)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c("NATIVE COVER" = "#1D9E75",
                              "NON-NATIVE COVER" = "#D85A30")) +
```



```

scale_y_continuous(labels = scales::percent) +
labs(x = "Survey year",
     y = "Proportion of vegetation cover",
     fill = "Cover category") +
theme_minimal() +
theme(legend.position = "bottom")

```

fig2

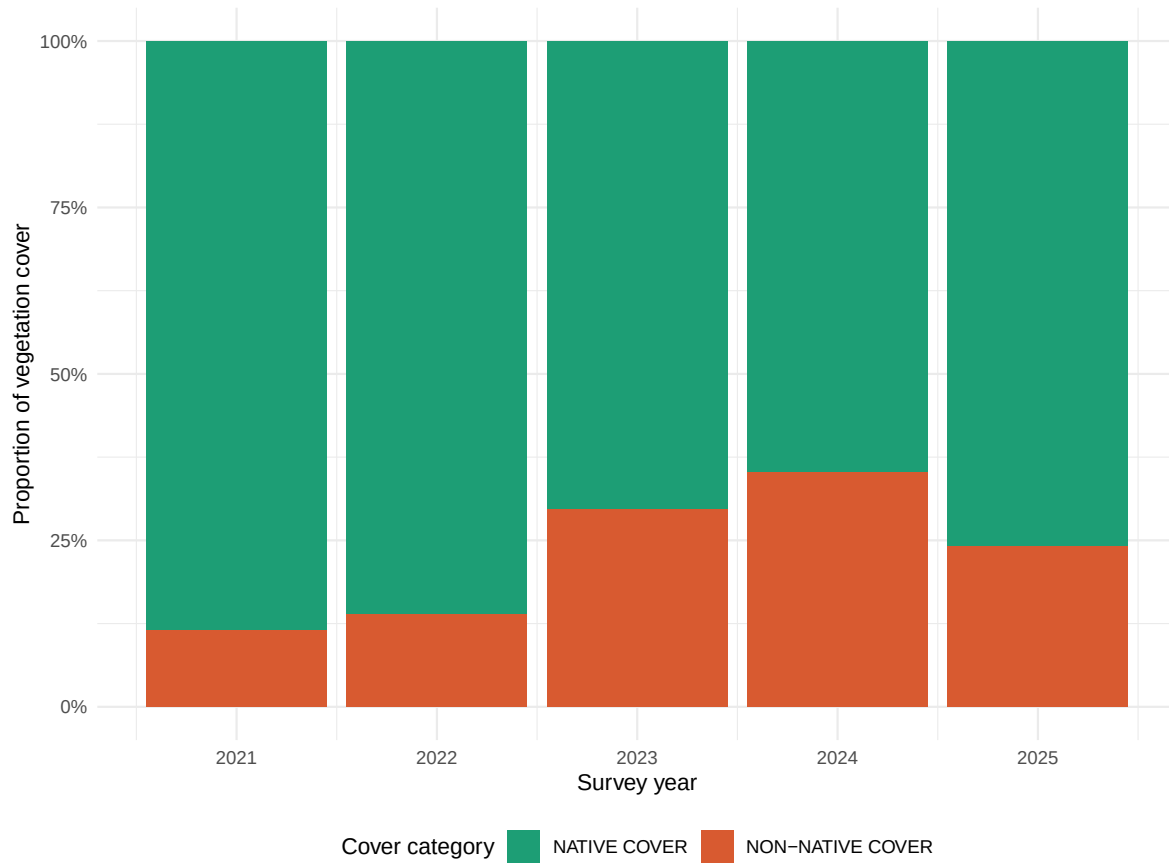


Figure 2: Proportional contribution of native and non-native vegetation cover in salt marsh habitat at NCOS across survey years. Each bar sums to 1. Native cover is shown in green; non-native cover in coral.

Figure 3: Species rank plot

```
# fig3 - pointrange instead of col
cutoff_x <- mean(c(species_rank$mean_pc[5], species_rank$mean_pc[6]))

fig3 <- ggplot(data = species_rank_top20,
               aes(x = mean_pc,
                   y = reorder(psoc, mean_pc),
                   color = cover_category)) +
  geom_pointrange(aes(xmin = mean_pc - se_pc,
                      xmax = mean_pc + se_pc)) +
  geom_vline(xintercept = cutoff_x,
             linetype = "dashed",
             color = "gray50") +
  scale_color_manual(values = c("NATIVE COVER" = "#1D9E75",
                                "NON-NATIVE COVER" = "#D85A30")) +
  labs(x = "Mean percent cover ( $\pm$  1 SE)",
       y = "Species",
       color = "Cover category") +
  theme_minimal() +
  theme(axis.text.y = element_text(face = "italic", size = 9),
        legend.position = "bottom",
        legend.title = element_text(face = "bold"))

fig3
```

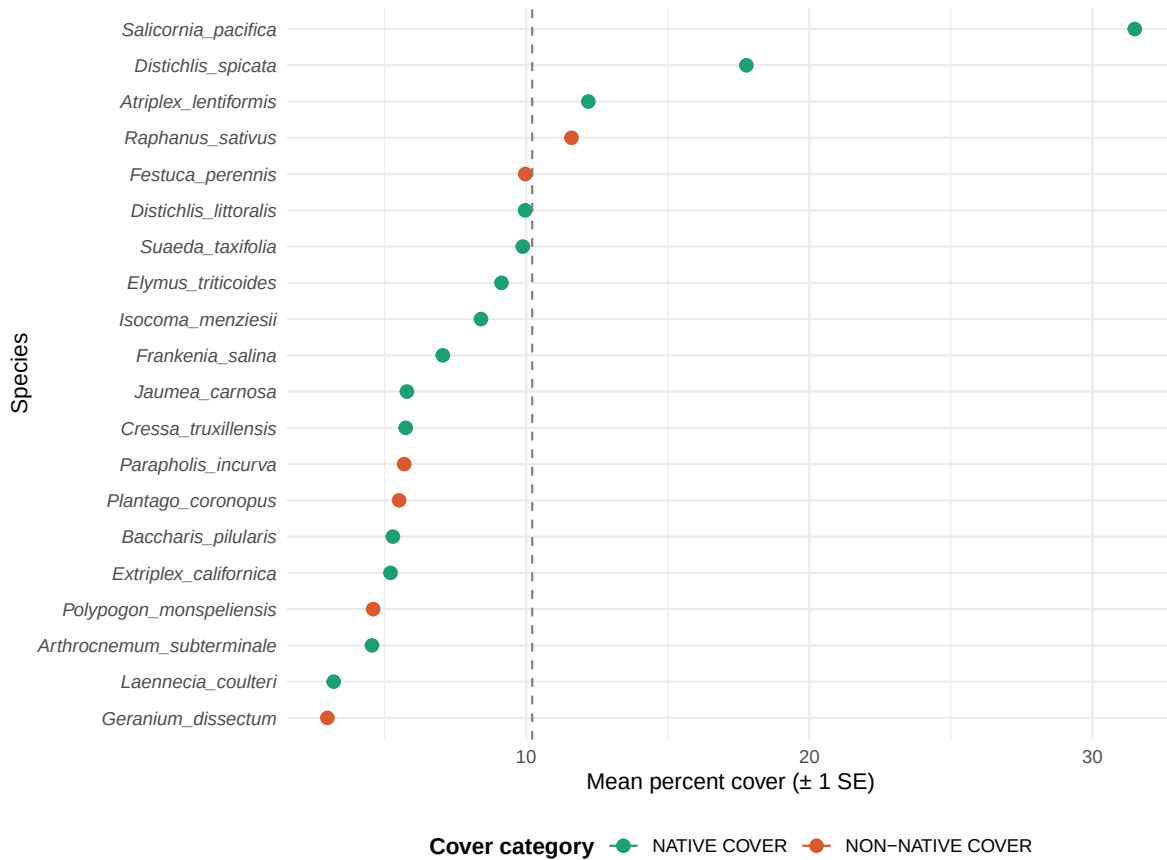


Figure 3: Mean percent cover (± 1 SE) of the top 20 plant species recorded in salt marsh habitat at NCOS, averaged across transects and survey years and ranked from highest to lowest mean cover. Points represent means and horizontal bars represent ± 1 standard error. Green indicates native species; coral indicates non-native species. Dashed line marks the cutoff for the five dominant species visualized in figure 1.

Figure 4: Total vegetation cover across survey years

```
fig4 <- ggplot(data = total_cover,
               aes(x = year,
                   y = mean_total)) +
  geom_ribbon(aes(ymin = mean_total - se_total,
                  ymax = mean_total + se_total),
             fill = "#1D9E75",
             alpha = 0.2) +
```

```
geom_line(color = "#1D9E75",
          linewidth = 1) +
geom_point(color = "#1D9E75",
           size = 3) +
labs(x = "Survey year",
     y = "Total mean percent cover (%)") +
theme_minimal()
```

fig4



Figure 4: Total mean percent vegetation cover (native + non-native combined) in salt marsh habitat at NCOS across survey years. Shaded band represents ± 1 standard error around the mean.